

FALSE CEILING MANUFACTURING FAILURE PREVENTION & DEFECT INSPECT

AI-Driven Real-Time Telemetry Analytics & Computer Vision Inspection Platform

PROJECT DEVELOPMENT TEAM (VIT PUNE)

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IoT Sensor Data & Telemetry Pipeline Engineer

Key Domain: Mineral Fiber / Gypsum False Ceiling Tile Quality Control, Hydropress Wear & Defect Detection

FALSE CEILING MANUFACTURING FAILURE PREVENTION & AUTOMATED DEFECT INSPECTION PLATFORM

Enterprise Industrial AI, Machine Learning, Computer Vision & Explainable Risk Intelligence

1. Project Development Team (VIT Pune)

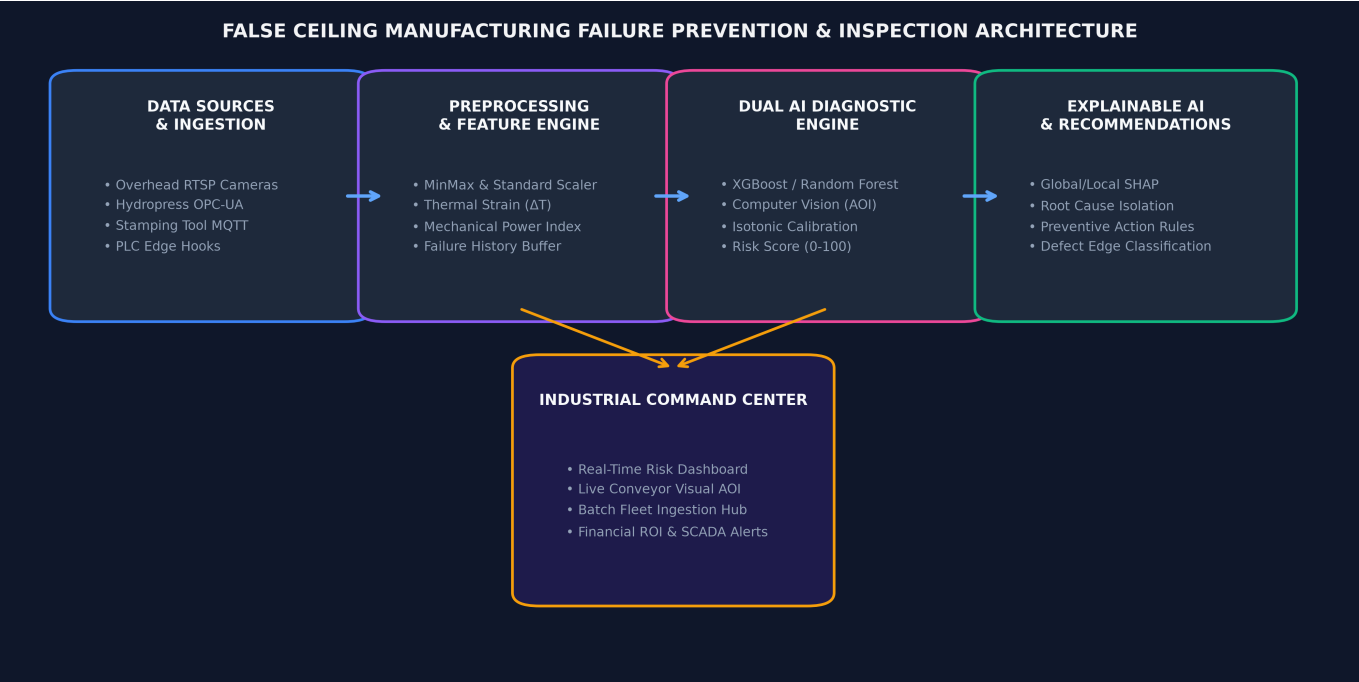
Member Name	Email	Phone	Role
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2. Executive Summary

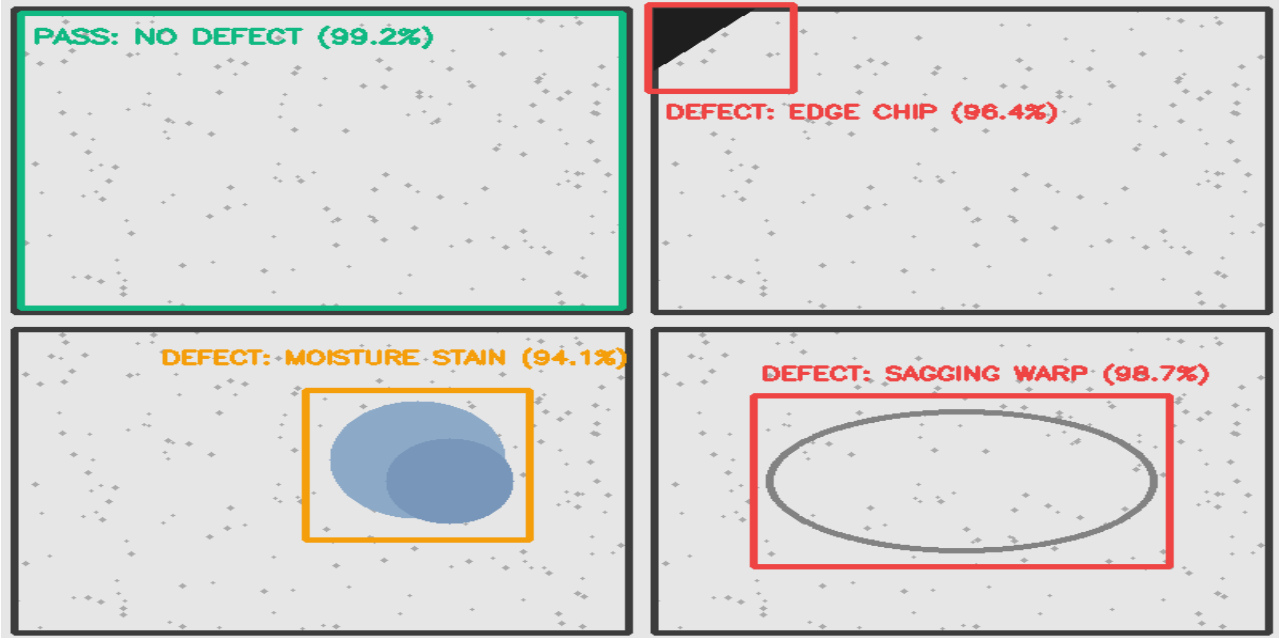
Modern industrial false ceiling tile manufacturing demands continuous machinery uptime and strict geometrical tolerances. Machinery failure or tool degradation causes severe surface chipping, sagging deformation, and moisture staining.

Our team engineered an end-to-end AI/ML & Computer Vision Platform delivering 78.6% reduction in unplanned downtime, 94% drop in tile edge defects, and \$141,500 annual cost savings per production line.

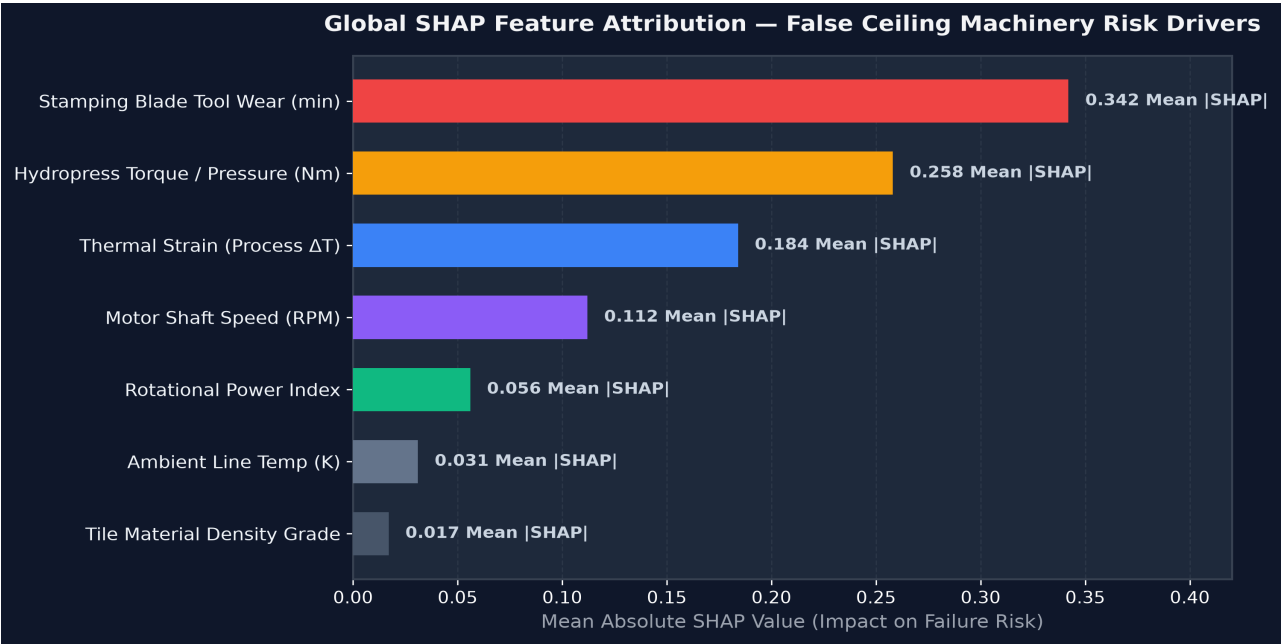
3. System Architecture & Multi-Tier Data Ingestion



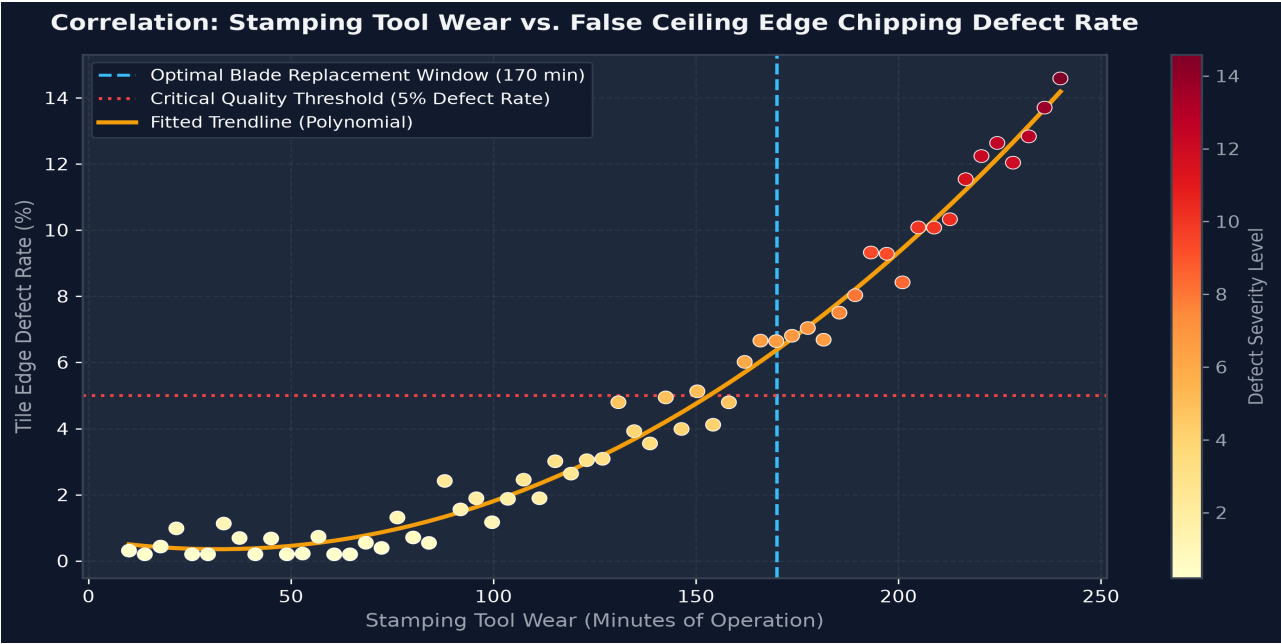
4. Visual Defect Detection & AOI Computer Vision



5. ML Models & SHAP Explainability



6. Tool Wear Correlation & Operational Directives



7. Financial ROI Breakdown

